

CHRONIC WOUNDS AND IMPAIRED HEALING

The previous discussion has focused on cutaneous wound repair following acute injury. However, the applicability of these processes is limited in the context of chronic wounds, such as those associated with diabetes, venous and arterial insufficiency, and conditions complicated by persistent inflammation or dysfunctional host responses. In chronic wounds, the linear, sequential progression through the phases of healing—hemostasis, inflammation, proliferation, and remodeling—is disrupted. There is no orderly transition between phases, and different areas of the same wound may be in different stages of repair simultaneously.

Acute wounds, such as surgical or traumatic injuries, follow a predictable healing timeline and typically resolve without complication. In contrast, chronic wounds are characterized by a failure to progress through the normal stages of healing, resulting in prolonged non-healing or impaired healing. Despite numerous observations and proposed hypotheses, studying the pathophysiology of chronic wounds remains challenging, primarily due to the lack of animal models that accurately replicate human impaired healing.

Some chronic wounds arise from a combination of ischemia, pressure, and infection. Diabetic foot ulcers exemplify this pathophysiologic triad. While the direct role of hyperglycemia in ulcer development remains debated, it is well established that hyperglycemia impairs neutrophil function, increasing susceptibility to infection in diabetic patients. Additionally, "small vessel disease" in diabetes is now understood not as an obstructive process but as a functional microvascular abnormality. Revascularization of the diabetic foot is considered appropriate when large-vessel disease is present and distal perfusion is adequate.

Venous ulcers represent another example of impaired healing, independent of pressure or arterial insufficiency. The primary underlying defect is venous

hypertension—defined as the failure of venous pressure in the lower extremities to decrease appropriately during muscle contraction and exercise. Several mechanisms have been proposed to explain how venous hypertension leads to ulceration or lipodermatosclerosis.

One hypothesis emphasizes the perivascular accumulation of fibrin due to impaired fibrinolysis, forming a "fibrin cuff" around dermal capillaries that may hinder nutrient and oxygen exchange. Another theory focuses on microvascular endothelial damage, leading to leakage of fibrinogen and other macromolecules into the dermis. This leakage may trap essential growth factors and signaling molecules, rendering them inaccessible for tissue repair or maintenance. Specifically, growth factors such as PDGF can be sequestered by leaked proteins like albumin, fibrinogen, and α_2 -macroglobulin—a known scavenger of growth factors.

Matrix metalloproteinases (MMPs) play a crucial role in normal wound repair, particularly in keratinocyte migration and extracellular matrix (ECM) remodeling. However, in chronic wounds, including venous ulcers, wound fluid contains elevated levels of MMPs. These excessive proteolytic enzymes degrade the ECM, as well as cytokines and growth factors, further impairing the healing process.

It is also important to recognize that the tissue surrounding chronic wounds is pathologically altered. These changes are driven by the underlying disease process or by the persistent failure to heal. A prominent clinical example is the fibrotic tissue surrounding venous ulcers, known as lipodermatosclerosis. Once established, lipodermatosclerosis creates a hostile environment for healing and serves as a common site for ulcer recurrence. Venous ulcers occurring within areas of lipodermatosclerosis are particularly difficult to treat and often require comprehensive management strategies targeting both the ulcer and the surrounding dysfunctional tissue.